SALMAN MOHAMMED ARIF

Data Science | Machine Learning | Applied AI | Python

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate focused on Data Science, Machine Learning, Applied AI, and Python development, with hands-on experience building AI-powered and data-driven applications. Skilled in Python, ML, NLP, FastAPI, and real-time APIs, with experience in technical leadership and hackathons. Seeking internship and entry-level opportunities to apply my skills to real-world AI and software engineering problems.

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL

Data Science: Pandas, NumPy, Data Analysis, Matplotlib, Seaborn

Machine Learning: Scikit-learn, TensorFlow, Random Forest, LSTM, Model Evaluation, Neural Networks

AI / NLP: Natural Language Processing, RAG Concepts, Speech-to-Text, AI-Assisted Data Processing

Backend & APIs: FastAPI, Flask, REST APIs, WebSockets

Databases: MySQL, SQLite, MongoDB

Tools: Git, GitHub, Visual Studio Code

EXPERIENCE

Software Engineering Intern - CodTech IT Solutions **May 2025 – June 2025**

- Built real-time web applications on a WebSocket architecture instead of standard polling, so backend and client stayed in sync without repeated request overhead.
- Integrated a third-party weather API into the backend, handling request failures and inconsistent response formats so the rest of the application could rely on clean, predictable data.
- Designed asynchronous backend logic to keep the application responsive under concurrent real-time connections rather than blocking on I/O-heavy operations.
- Worked across the stack - from external API integration to backend services to live data delivery - gaining a full picture of how real-time systems are structured end to end.

PROJECTS

JourneyIt - AI-Powered Travel Intelligence Platform

- Designed a full-stack travel intelligence platform covering flight-price prediction, fare trend analysis, and booking recommendations - owning the pipeline from raw API data to user-facing insight.
- Combined LSTM and Random Forest models deliberately: LSTM to capture sequential fare trends over time, Random Forest to handle non-sequential features like route and carrier, then blended both into a single forecast.
- Integrated multiple live flight and travel APIs (AviationStack, Booking.com) and normalized their differing data formats to support consistent price ranking and recommendation logic.
- Built an AI chatbot layer on top of the forecasting engine so users could query fare trends and get recommendations in natural language instead of navigating raw data.

Technologies: *Python, FastAPI, React, TensorFlow, Scikit-learn, Firebase, Gemini API, AviationStack API, Booking.com API*

AgriSathi - USSD-Based Smart Agriculture Advisory System

- Designed the system around USSD rather than a smartphone app specifically so farmers on basic feature phones, without internet access, could still reach crop prices, weather updates, and farming guidance.

- Combined rule-based logic for straightforward queries with data-driven processing for more variable ones, balancing reliability with flexibility on a constrained USSD interface, and added SMS as a fallback channel.
- Architected the system to be extended later with AI-based recommendations and voice assistance, so accessibility improvements could be layered in without rebuilding the core advisory logic.

Technologies: *Python, Flask, USSD, SMS, Data APIs, React Native*

BizLedger - AI-Powered Voice Transaction Recording App

- Built a voice-to-ledger pipeline that turns unstructured spoken input into structured financial records, removing the need for manual data entry in small-business bookkeeping.
- Applied speech-to-text transcription followed by NLP-based information extraction to pull transaction details (amount, party, category) out of free-form speech.
- Automated SQL generation from extracted transaction data so records could be written directly to the database without a manual mapping step between input and storage.

Technologies: *Python, FastAPI, Speech-to-Text, NLP, SQL, React Native*

EDUCATION

B.E. - Computer Science and Engineering

Muffakham Jah College of Engineering and Technology (MJCET) | 2022 – 2026 | CGPA: 8.2 / 10

XII - Higher Secondary Education

Narayana Institute | 2020 – 2022 | Score: 948 / 1000

CERTIFICATIONS

- Python Essentials 1 - Cisco Networking Academy (August 2026)
- AWS Cloud Practitioner Essentials - AWS Training & Certification (July 2026)
- AI Upskilling Certificate: Technical Foundation - Qualcomm Academy (January 2026)
- The Ultimate Job Ready Data Science Course - CodeWithHarry (January 2026)

LEADERSHIP & ACHIEVEMENTS

- Logistics Head, Google Developer Groups on Campus (GDGC), MJCET
- Marketing Head, IEEE Student Branch, MJCET
- Team Lead, DATANYX 24-Hour Datathon, MJCET
- Advisor - Sponsorship & PR, Computer Society of India (CSI), MJCET
- Branch Sports Coordinator, CSE Department, MJCET
- Manager, MJ Esports Community, MJCET
- Sports Secretary, Sunflower School
- Participant, Hack Revolution, MJCET
- 3rd Place, Map Marking Competition, International Indian School Jeddah (IISJ)

PROFESSIONAL DEVELOPMENT & RESEARCH

- Preparing for AWS Certified Cloud Practitioner (CLF-C02) examination
- Preparing for GitHub Foundations examination
- Research Paper - AgriSathi: development in progress

LANGUAGES

English, Arabic, Hindi, Urdu